

Perrys Handbook Heat Transfer Fluid Flow

Page 1: Fluid Flow Calculations and Head Loss

Perry's Chemical Engineers' Handbook outlines the equations for fluid flow. Pressure drop due to friction in straight pipes is calculated using the Darcy-Weisbach equation: $h_f = f \cdot (L / D) \cdot (v^2 / (2 \cdot g))$, where h_f is friction head loss, f is the friction factor, L is pipe length, D is inside diameter, v is velocity, and g is gravitational acceleration.

The friction factor f is determined based on the Reynolds number ($Re = D \cdot v \cdot \rho / \mu$). For laminar flow ($Re < 2100$), $f = 64 / Re$. For turbulent flow ($Re > 4000$), f is determined using the Colebrook equation or the Moody diagram based on pipe relative roughness.

Page 2: Convective and Conductive Heat Transfer

Heat transfer by conduction is governed by Fourier's Law: $q = -k * A * (dT/dx)$, where k is the thermal conductivity of the material. Convective heat transfer is governed by Newton's Law of Cooling: $q = h * A * (T_s - T_f)$, where h is the convective heat transfer coefficient.

In shell-and-tube heat exchangers, the overall heat transfer coefficient U is calculated as: $1/U = 1/h_o + R_{fo} + (d_o/d_i)*(1/h_i) + R_{fi} + R_w$, where h_o and h_i are shell and tube side film coefficients, R_{fo} and R_{fi} are fouling factors, and R_w is the tube wall resistance.